

Characteristic Domination for General L^q -Valued Local Martingales

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Status

This packet gives a candidate partial result, likely valid, for Remark 12.10 of Yaroslavtsev’s paper *Local characteristics and tangency of vector-valued martingales* [1]. It proves the full general-local-martingale characteristic-domination conclusion for $X = L^q(S)$, $1 < q < \infty$, and $1 < p < \infty$.

It is partial relative to the source’s broad UMD-space problem because it only handles L^q -spaces, and it does not cover the endpoint $p = 1$.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. After proving the quasi-left-continuous characteristic-domination theorem, Yaroslavtsev states that the general local-martingale case is open and identifies the missing issue as a discrete independent-increment comparison.

... (12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The continuation formulates the independent mean-zero aggregate-domination problem and notes that even the symmetric case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

Main Result

Theorem 1. *Let $1 < p, q < \infty$, let (S, Σ, ρ) be a measure space, and put $X = L^q(S)$. Let N and M be X -valued local martingales. Assume that N is characteristically dominated by M in the sense of [1, Definition 12.9]: for every $x^* \in X^*$,*

$$|\langle N_0, x^* \rangle| \leq |\langle M_0, x^* \rangle|, \quad \llbracket N^c \rrbracket_\infty \leq \llbracket M^c \rrbracket_\infty,$$

and

$$\nu^N(\mathbb{R}_+ \times B) \leq \nu^M(\mathbb{R}_+ \times B) \quad (B \in \mathcal{B}(X), 0 \notin B)$$

almost surely. Then

$$\mathbb{E} \sup_{t \geq 0} \|N_t\|_{L^q(S)}^p \leq C_{p,q} \mathbb{E} \sup_{t \geq 0} \|M_t\|_{L^q(S)}^p.$$

Idea of the Proof

The earlier discrete packet proved the independent-sum bottleneck by using the sharp L^q -valued Rosenthal theorem of Dirksen and Yaroslavtsev [2]. The general local case has one extra trap: characteristic domination compares only the total spatial jump compensators $\nu(\mathbb{R}_+ \times \cdot)$. It does not compare the accessible and quasi-left-continuous jump components separately.

The way around this is to show that, in L^q , the relevant Rosenthal sum/intersection norm of the identity jump x depends only on the total spatial compensator. Time labels and the accessible/qlc tag can be erased. For the basic S , $D_{q,q}$, and $D_{p,q}$ gauges this is immediate from the formulas. For sum spaces, a time-dependent split can be conditionally averaged over the spatial jump value x ; Jensen's inequality decreases all three gauges. Thus the jump part has a single monotone norm controlled by $\nu(\mathbb{R}_+ \times \cdot)$. The continuous part is already controlled by Yaroslavtsev's continuous characteristic-domination estimate.

The Total Jump Rosenthal Norm

Let $\bar{\nu} = \bar{\nu}_\omega$ be a predictable random measure on $X \setminus \{0\}$, where $X = L^q(S)$. For a measurable $F : \Omega \times X \rightarrow X$, define

$$\|F\|_{S(\bar{\nu})} = \left(\mathbb{E} \left\| \left(\int_X |F(\omega, x)|^2 \bar{\nu}_\omega(dx) \right)^{1/2} \right\|_{L^q(S)}^p \right)^{1/p},$$

$$\|F\|_{D_r(\bar{\nu})} = \left(\mathbb{E} \int_X \|F(\omega, x)\|_{L^q(S)}^r \bar{\nu}_\omega(dx) \right)^{1/r} \quad (r = p),$$

and

$$\|F\|_{D_q(\bar{\nu})} = \left(\mathbb{E} \left(\int_X \|F(\omega, x)\|_{L^q(S)}^q \bar{\nu}_\omega(dx) \right)^{p/q} \right)^{1/p}.$$

Let $R_{p,q}(\bar{\nu})$ be the norm of the identity map $I(\omega, x) = x$ in the same six sum/intersection spaces as in the Dirksen–Yaroslavtsev theorem:

$$\begin{aligned} S \cap D_q \cap D_p, & \quad 2 \leq q \leq p < \infty, \\ S \cap (D_q + D_p), & \quad 2 \leq p \leq q < \infty, \\ (S \cap D_q) + D_p, & \quad 1 < p < 2 \leq q < \infty, \\ (S + D_q) \cap D_p, & \quad 1 < q < 2 \leq p < \infty, \\ S + (D_q \cap D_p), & \quad 1 < q \leq p \leq 2, \\ S + D_q + D_p, & \quad 1 < p \leq q \leq 2. \end{aligned}$$

Intersections carry the maximum norm, and sums carry the usual infimum norm.

Collapse and Monotonicity

Lemma 1. *Let $\kappa_\omega(dt, dx)$ be a predictable random measure on $\mathbb{R}_+ \times X$, and let*

$$\bar{\kappa}_\omega(B) = \kappa_\omega(\mathbb{R}_+ \times B)$$

be its total spatial marginal. The Rosenthal sum/intersection norm of the identity jump x , computed with respect to κ , equals the same norm computed with respect to $\bar{\kappa}$. In particular it is insensitive to the time distribution of the compensator.

Moreover, if $\bar{\nu}_\omega \leq \bar{\mu}_\omega$ almost surely, then

$$R_{p,q}(\bar{\nu}) \leq R_{p,q}(\bar{\mu}).$$

Proof. For the basic gauges, the first assertion is just Fubini:

$$\int_{\mathbb{R}_+ \times X} |x|^2 \kappa_\omega(dt, dx) = \int_X |x|^2 \bar{\kappa}_\omega(dx),$$

and similarly for $\|x\|_{L^q}^p$ and $\|x\|_{L^q}^q$.

The only point is a sum norm, where one may have allowed a decomposition

$$x = F_1(\omega, t, x) + \cdots + F_m(\omega, t, x)$$

depending on time. Disintegrate $\kappa_\omega(dt, dx)$ over its x -marginal and define

$$G_j(\omega, x) = \mathbb{E}_\kappa(F_j(\omega, t, x) \mid \omega, x).$$

Then $x = \sum_j G_j(\omega, x)$. Since $u \mapsto |u|^2$ in the lattice order and $u \mapsto \|u\|_{L^q}^r$ for $r = p, q$ are convex, Jensen's inequality gives

$$\int |G_j|^2 d\bar{\kappa} \leq \int |F_j|^2 d\kappa, \quad \int \|G_j\|_{L^q}^r d\bar{\kappa} \leq \int \|F_j\|_{L^q}^r d\kappa.$$

After taking the outer $L^p(\Omega)$ norms, no basic gauge increases. Thus time-dependent decompositions give no smaller sum/intersection norm than spatial decompositions. The reverse inequality is obtained by viewing any spatial decomposition as time-independent.

For monotonicity, the three basic gauges are monotone under measure domination. For sums, argue recursively. Suppose a sum expression $E + F$ is under consideration and choose an almost optimal split $I = U + V$ for the larger measure $\bar{\mu}$. Let $h = d\bar{\nu}/d\bar{\mu}$, $0 \leq h \leq 1$. Restrict the split measure of (x, U, V) by the weight $h(\omega, x)$. This gives a split of the identity under $\bar{\nu}$, and

the aggregate measures of the two pieces are dominated by the corresponding pieces for $\bar{\mu}$. The recursive hypothesis bounds the E - and F -norms of the restricted pieces by those of U and V . Letting the near-optimality error tend to zero gives the sum step. Intersections are immediate from the maximum norm.

The argument is first written for simple finite-range decompositions. The general case follows by the standard separable-range and monotone/density approximations used in the Dirksen–Yaroslavl'tsev spaces. $\square$

Corollary 1. *Let L be an $L^q(S)$ -valued local martingale with $L_0 = 0$, and let $L^d = L^q + L^a$ be the purely discontinuous part of its canonical decomposition. Then, for $1 < p, q < \infty$,*

$$\left(\mathbb{E} \sup_{t \geq 0} \left\| L_t^d \right\|_{L^q(S)}^p \right)^{1/p} \simeq_{p,q} R_{p,q}(\nu^L(\mathbb{R}_+ \times \cdot)).$$

Proof. Dirksen and Yaroslavl'tsev prove sharp estimates for both jump types. For the quasi-left-continuous part, their non-atomic random-measure theorem identifies the maximal $L^p(\Omega; L^q(S))$ norm of $\int x d\bar{\mu}^{L^q}$ with the $\mathcal{I}_{p,q}$ -norm computed from the compensator ν^{L^q} . For the accessible part, their accessible-jump theorem identifies the maximal norm with the accessible $\mathcal{A}_{p,q}$ -norm, whose defining predictable square and D -gauges are exactly the same integrals against the accessible compensator ν^{L^a} .

Their combined general-local estimate, applied with continuous part zero, gives

$$\left\| L^d \right\|_{p,*} \simeq_{p,q} \|x\|_{\mathcal{I}_{p,q}(\nu^{L^q})} + \|L^a\|_{\mathcal{A}_{p,q}},$$

where $\|L\|_{p,*} := (\mathbb{E} \sup_t \|L_t\|_{L^q}^p)^{1/p}$. By Lemma 1, the right side is equivalent to the single sum/intersection norm of the identity with respect to the total spatial measure

$$\nu^{L^q}(\mathbb{R}_+ \times \cdot) + \nu^{L^a}(\mathbb{R}_+ \times \cdot) = \nu^L(\mathbb{R}_+ \times \cdot).$$

This proves the claim. $\square$

Proof of Theorem 1

It suffices to treat martingales starting at zero; the initial values are handled at the end. Since $L^q(S)$ is UMD for $1 < q < \infty$, Yaroslavl'tsev's canonical decomposition theorem [1, Theorem 3.5] applies. Write

$$M = M^c + M^d, \quad N = N^c + N^d,$$

where $M^d = M^q + M^a$ and $N^d = N^q + N^a$.

For the continuous terms, Yaroslavl'tsev's continuous characteristic-domination estimate, used in the source proof of the quasi-left-continuous domination theorem, gives

$$\left(\mathbb{E} \sup_t \|N_t^c\|_{L^q}^p \right)^{1/p} \lesssim_{p,q} \left(\mathbb{E} \sup_t \|M_t^c\|_{L^q}^p \right)^{1/p}.$$

For the discontinuous terms, Corollary 1 and the total compensator domination give

$$\begin{aligned} \left(\mathbb{E} \sup_t \left\| N_t^d \right\|_{L^q}^p \right)^{1/p} &\simeq_{p,q} R_{p,q}(\nu^N(\mathbb{R}_+ \times \cdot)) \\ &\leq R_{p,q}(\nu^M(\mathbb{R}_+ \times \cdot)) \\ &\simeq_{p,q} \left(\mathbb{E} \sup_t \left\| M_t^d \right\|_{L^q}^p \right)^{1/p}. \end{aligned}$$

Notice that no comparison of accessible parts with accessible parts, or quasi-left-continuous parts with quasi-left-continuous parts, is used here. The comparison is made only after the two jump types have been folded into the total spatial compensator.

Combining the continuous and discontinuous estimates, and using the strong L^p equivalence for the canonical decomposition [1, Theorem 3.5], yields

$$\begin{aligned} \left(\mathbb{E} \sup_t \|N_t\|_{L^q}^p \right)^{1/p} &\leq \left(\mathbb{E} \sup_t \|N_t^c\|_{L^q}^p \right)^{1/p} + \left(\mathbb{E} \sup_t \|N_t^d\|_{L^q}^p \right)^{1/p} \\ &\lesssim_{p,q} \left(\mathbb{E} \sup_t \|M_t^c\|_{L^q}^p \right)^{1/p} + \left(\mathbb{E} \sup_t \|M_t^d\|_{L^q}^p \right)^{1/p} \\ &\lesssim_{p,q} \left(\mathbb{E} \sup_t \|M_t\|_{L^q}^p \right)^{1/p}. \end{aligned}$$

If the martingales do not start at zero, the hypothesis on initial values implies $\|N_0\|_{L^q} \leq \|M_0\|_{L^q}$. Apply the zero-start result to $N - N_0$ and $M - M_0$, and use

$$\sup_t \|N_t\|_{L^q} \leq \|N_0\|_{L^q} + \sup_t \|N_t - N_0\|_{L^q},$$

together with the analogous bound for M . This gives the stated inequality.

Verification Notes

The proof was checked against the source passages:

- Yaroslavl'tsev's canonical-decomposition theorem: canonical decomposition and strong maximal L^p equivalence in UMD spaces.
- Yaroslavl'tsev's decoupled-tangent and uniqueness results: independent increments given local characteristics, used as background for Remark 12.10's reduction.
- Yaroslavl'tsev [1, Definition 12.9 and Remark 12.10]: characteristic domination compares the continuous covariation and the total spatial jump compensator.
- Dirksen–Yaroslavl'tsev's accessible-jump, non-atomic random-measure, and combined general-local estimates: sharp L^q -valued estimates for the jump parts used in Corollary 1.

The main audit point is Lemma 1. It is the step that prevents the false componentwise comparison $\nu^{N^a} \leq \nu^{M^a}$, which characteristic domination does not provide.

Limitations and Novelty Check

The result covers arbitrary measure spaces S , all $1 < q < \infty$, and all $1 < p < \infty$. It does not cover $p = 1$, because the sharp Dirksen–Yaroslavl'tsev L^q Rosenthal estimates used here are stated for $1 < p < \infty$. It does not settle the source question for arbitrary UMD spaces.

Cheap run-index searches found earlier packets for c_0 , Hilbert spaces, ℓ_q at the natural exponent, and the discrete L^q aggregate comparison, but no packet proving the full L^q general-local conclusion. The literature checked for this push included arXiv:1907.11588, arXiv:1707.00109, and nearby BDG/local-characteristics material. I found the ingredients above, but not an explicit statement of this total-compensator L^q consequence.

Human Review Recommendation

High priority. If Lemma 1 is accepted, the proof appears to close the full $L^q(S)$, $1 < p, q < \infty$, general-local-martingale subcase of Remark 12.10.

References

- [1] I. S. Yaroslavtsev, *Local characteristics and tangency of vector-valued martingales*, arXiv:1907.11588.
- [2] S. Dirksen and I. S. Yaroslavtsev, *L^q -valued Burkholder–Rosenthal inequalities and sharp estimates for stochastic integrals*, arXiv:1707.00109.